

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 13-2023: A 25-Year-Old Woman with Abdominal Pain and Jerking Movements

Leigh H. Simmons, M.D., Mladen Nisavic, M.D., and Amy K. Dickey, M.D.

PRESENTATION OF CASE

Dr. John C. Schell (Medicine): A 25-year-old woman with a history of acute intermittent porphyria was admitted to this hospital because of abdominal pain and jerking movements of the right arm and leg.

The patient had been in her usual state of health until 2 days before this admission, when diffuse abdominal pain and nausea developed. She also felt confused and agitated, and a rash developed on her chest. One day before this admission, four episodes of jerking movements of the right arm and leg occurred, and the patient was concerned that she was having seizures. She presented to another hospital for evaluation.

At the other hospital, the patient reported that the symptoms were consistent with her previous flares of acute intermittent porphyria. She reported that jerking movements of the right arm and leg had occurred 5 minutes before her arrival at the hospital and had caused jaw dislocation; she also reported that the rash on her chest had become painful. On examination, the patient appeared uncomfortable. The mental-status and neurologic examinations were reportedly normal. The jaw was dislocated to the left, and there was diffuse abdominal tenderness. Intravenous morphine, diphenhydramine, and ondansetron were administered; an infusion of dextrose was started; and the dislocated jaw was manually reduced. The patient was admitted to the other hospital. The next day, the severity of pain in the abdomen and rash increased. Transfer to this hospital was requested because of a shortage of hemin at the other hospital.

On evaluation at this hospital, the patient described intense pain in the abdomen, a painful rash, and persistent nausea, agitation, and confusion. She was worried that she might have seizures, have blistering of the skin, or die because the hemin infusion had been delayed. Additional history was obtained from the patient. She had a history of obesity, anxiety, and chronic pain that had been attributed to acute intermittent porphyria. She reported that acute intermittent porphyria had been diagnosed 13 years before this admission, when abdominal pain, jerking movements of the right arm and leg, and a painful rash developed during menarche. During the 13 years before this admission, one or two flares of

From the Departments of Medicine (L.H.S., A.K.D.) and Psychiatry (M.N.), Massachusetts General Hospital, and the Departments of Medicine (L.H.S., A.K.D.) and Psychiatry (M.N.), Harvard Medical School — both in Boston.

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acute intermittent porphyria had occurred each year; during the 6 months before this admission, no flares had occurred.

Two previous flares of acute intermittent porphyria had been treated with a hemin infusion. During the first hemin infusion, the patient had a reaction that was characterized by intense yelling and writhing in pain, and intubation was performed to complete the infusion. Before the second hemin infusion, she received intravenous methylprednisolone and underwent elective intubation to prevent a reaction. The patient identified heat, stress, and menses as triggers for acute intermittent porphyria flares.

Medications included intravenous diphenhydramine and dextrose, which were administered daily through a central-venous-access port in the right side of the chest. The patient had previously taken intravenous morphine daily; however, she had not had access to this medication for approximately 3 months because of an inability to transport it from her home in the United Kingdom. Nonsteroidal antiinflammatory drugs and penicillin had caused anaphylaxis, prochlorperazine had caused flushing, metoclopramide had caused anxiety, and fentanyl had caused altered mental status.

The patient was born in New England and had moved to the United Kingdom 15 years earlier. She was a student in a graduate program at a university in the United Kingdom and had returned to New England to visit her family. She did not use alcohol or illicit drugs, and she did not vape or smoke cigarettes. Her mother had dementia, and her father had died years earlier from an unknown cause. Her maternal grandmother had acute intermittent porphyria. Her immediate family had undergone genetic testing for acute intermittent porphyria; her brother was an asymptomatic carrier of a pathogenic variant, and her six other siblings were not carriers.

On examination, the temperature was 36.3°C, the blood pressure 144/90 mm Hg, the heart rate 100 beats per minute, the respiratory rate 18 breaths per minute, and the oxygen saturation 97% while the patient was breathing ambient air. She appeared older than her stated age. She was awake, alert, and oriented. She was frustrated and irritable, with pressured speech. A port in the right side of the chest was palpated. Tenderness in the abdomen was diffuse but was

worst in the right upper quadrant. Erythematous plaques and excoriations were present on the arms and chest and under the breasts. The neurologic examination was normal.

Blood levels of electrolytes, calcium, total protein, albumin, and globulin were normal, as were the results of kidney-function and liver-function tests. The hemoglobin level was 10.6 g per deciliter (reference range, 12.0 to 16.0) and the platelet count 120,000 per microliter (reference range, 150,000 to 400,000). A urine specimen was obtained for measurement of the urinary porphobilinogen (PBG) and porphyrin levels, but the amount of urine was inadequate. Oral acetaminophen and intravenous morphine, diphenhydramine, lorazepam, and ondansetron were administered. Infusions of dextrose and hemin were started, and the patient was admitted to this hospital.

During the subsequent 2 days, the hemin infusion was continued, but the patient's symptoms did not abate. On the second hospital day, an additional urine specimen was obtained; the urinary PBG level was 0.3 μ mol per liter (reference range, 0.0 to 1.3) and the uroporphyrin level 4 nmol per liter (reference range, 0 to 30). On the third hospital day, the patient reported worsening jerking movements of the right arm and leg, which she believed had caused jaw dislocation. On examination, the jaw was dislocated to the left. She did not cooperate with a manual reduction, and a surgical reduction of the jaw was performed.

A diagnosis was made.

DIFFERENTIAL DIAGNOSIS

Dr. Leigh H. Simmons: If I were hearing this case presented on morning rounds, a few things would catch my attention right away. I would be very concerned about the extent of this patient's illness. This young woman has been transferred to our hospital with a dramatic presentation, an apparent exacerbation of a serious and rare disease — one that I have not treated before, although I have considered the diagnosis many times in the evaluation of abdominal pain. She tells us she is afraid that she may have seizures or die from not receiving a hemin infusion promptly. There is an urgent need to figure out how to help her.

The key features of this patient's case include intermittent abdominal pain for 13 years, jerking movements on one side of the body, painful rashes, and a family history of acute intermittent porphyria with genetic testing of several family members. We are not given the genetic test results for the patient or her parents. It is also striking that the patient has managed this complex illness and its treatment in two different countries. She has received intravenous morphine through a central-venous-access port, although not during the 3 months before this presentation, and she reports a severe reaction to a hemin infusion. Recurrent jaw dislocation has also occurred.

My thoughts first turn to the patient's diagnosis of acute intermittent porphyria. Acute intermittent porphyria is a disease caused by heterozygosity for pathogenic variants in PBG deaminase, which leads to a buildup of heme precursors. Patients with symptomatic acute intermittent porphyria can have abdominal pain, arm and leg pain, paresthesia, weakness, and tachycardia. This patient reports that she is having a typical flare of acute intermittent porphyria. However, several features of her presentation are not consistent with this diagnosis. Her rashes are not typical of acute intermittent porphyria, although similar rashes have been associated with other forms of porphyria. Her pain and neurologic symptoms are not alleviated by standard treatment with hemin and dextrose infusions, nor by morphine. Whenever I evaluate a patient in whom a rare illness has been diagnosed, I always take a step back to confirm that I understand how the disease was diagnosed in the patient and determine whether the diagnosis is indeed correct. Therefore, the first step is to determine whether this patient has acute intermittent porphyria.

FLARE OF ACUTE INTERMITTENT PORPHYRIA

Given the patient's reported history of acute intermittent porphyria, the medical team was right to quickly treat her with dextrose and then transfer her to a hospital that had hemin. An untreated flare of acute intermittent porphyria can be permanently disabling or fatal. The team was also right to check the urinary PBG and porphyrin levels; it is important to confirm that the current symptoms are due to a flare, even in patients with established acute intermittent por-

phyria. This testing is relatively inexpensive, highly sensitive, and critical in the diagnosis of an acute intermittent porphyria flare.

In this patient, the diagnosis remains in question. Because she has an unusual course of illness and a poor response to standard treatment, I would involve a consultant with expertise in porphyria. I would ask the consultant two questions. First, is the patient's history of a severe pain reaction to a hemin infusion common? Second, how should we interpret the normal urinary PBG and porphyrin levels, knowing that she had been treated with hemin for 1 day at the time that these levels were obtained?

INTRAABDOMINAL INFLAMMATION OR INFECTION

Intraabdominal inflammation or infection is also a consideration. Her severe abdominal pain could be caused by bowel obstruction, biliary colic, appendicitis, or pancreatitis. However, no laboratory test results are provided that would support these diagnoses. Cross-sectional imaging would help to rule out or establish a diagnosis of intraabdominal inflammation or infection. I suspect that such imaging studies have been obtained during past attacks. If previous studies cannot be reviewed, computed tomography would probably be indicated.

MEDICATION OR TOXIN

Could this patient's jerking motions, which are possibly seizures, be caused by a medication or toxin? Toxic effects of medications or substances must be considered. A possible culprit medication is diphenhydramine, which in excess doses can cause myoclonus. Lead poisoning should also be considered in a patient with abdominal pain and neuropsychiatric symptoms. Laboratory data that would support this diagnosis include normocytic anemia, elevated aminotransferase levels, and basophilic stippling; the diagnosis could be easily established by obtaining a blood lead level.¹

In addition, I would give serious consideration to a withdrawal syndrome. This patient no longer has access to her usual morphine treatment. She has some symptoms that could be consistent with opiate withdrawal, including irritability, restlessness, and abdominal pain, but she notably does not have vomiting, diarrhea, or sweating on presentation. Also, her symptoms

are not alleviated by the morphine that she is receiving in the hospital, which makes an ongoing opiate withdrawal syndrome unlikely.

FACTITIOUS DISORDER

At this point, I am concerned that this patient's presentation may not be explained by any of these illnesses. Instead, the features of her presentation may be suggestive of factitious disorder. Factitious disorder is characterized by falsified illness and deceptive behavior regarding symptoms. This disorder may be suspected when a patient has been to multiple health care facilities, has an inconsistent medical history, does not allow access to collateral information from family and friends, and has symptoms that are not responding as expected to standard treatments.

Abdominal pain and seizures are among the symptoms that are most commonly reported by patients with factitious disorder. There is often a history of childhood illness; illness in family members is also common. A timely evaluation for factitious disorder is essential. In this case, the patient is receiving hemin treatment and other medical interventions that seem to be causing her harm. There was a hemin shortage at the time of her presentation, so treating her unnecessarily could have deprived other patients of an essential treatment.

I suspect that the diagnosis in this case was made in one of three ways. First, staff could have observed the patient doing something suspicious, such as dislocating her jaw, manipulating her port, or tampering with specimens. Second, a family member or friend could have provided collateral information that was suggestive of deception. Third, a review of the electronic health record, perhaps through a shared network, could have provided clues that the patient was deliberately causing illness. Cases in which the diagnosis of factitious disorder was discovered through a search of the electronic health record have been reported in the literature; the electronic health record can be a powerful tool to expedite the diagnosis and avoid unnecessary treatments for these patients.² In this case, my diagnostic approach would be a detailed interview of the patient regarding her illness history and a request for medical records.

Table 1. Testing for Acute Hepatic Porphyrria.

First-line testing
Measurement of urinary porphobilinogen
Measurement of urinary total porphyrins
Second-line testing
Measurement of urinary aminolevulinic acid
Plasma porphyrin analysis
Fecal porphyrin analysis
Plasma fluorescence scanning
Testing for porphobilinogen deaminase activity
Genetic testing

DR. LEIGH H. SIMMONS'S DIAGNOSIS

Factitious disorder.

EVALUATION FOR ACUTE HEPATIC PORPHYRIA

Dr. Amy K. Dickey: Unfortunately, laboratory test results needed to confirm or rule out acute hepatic porphyria (including acute intermittent porphyria) were not obtained before the administration of hemin in this patient. The urinary PBG and uroporphyrin levels were both normal 1 day after the initiation of treatment with hemin.

Testing for acute hepatic porphyria (Table 1) is performed when a patient has undiagnosed abdominal pain or nonspecific neurologic symptoms and the initial workup for more common causes is negative.³⁻⁵ A low threshold for testing is encouraged. First-line testing for acute hepatic porphyria involves measurement of the urinary PBG and porphyrin levels.^{4,5} It is notable that PBG is not a porphyrin and may not be included in porphyrin panels. Testing of random urine samples is recommended, and results should be normalized to the urinary creatinine level.

Although acute hepatic porphyria is probably underdiagnosed among some patients, the disease is overdiagnosed among other patients, particularly those who receive a diagnosis after detection of nonspecific elevated porphyrin levels but without measurement of the urinary PBG level. During acute attacks of porphyria, the urinary PBG level is expected to exceed 10 mg per gram of creatinine.^{4,5} Elevated porphyrin

levels are nonspecific in the absence of an elevated urinary PBG level or evidence of porphyria on additional testing. Porphyrin levels can be elevated under many circumstances, such as liver disease, heavy-metal poisoning, and the use of certain medications.^{4,5} However, first-line testing includes urinary porphyrin testing because porphyrin levels are typically elevated longer than the PBG level and can be useful when the PBG level is not markedly elevated or is normal after resolution of symptoms. Between acute attacks, biochemical intermediates may remain elevated or return to normal.

When first-line testing is positive, second-line testing is performed to confirm the diagnosis and differentiate the type of acute hepatic porphyria. Second-line testing includes measurement of the urinary aminolevulinic acid level, plasma and fecal porphyrin analysis, plasma fluorescence scanning, and testing for erythrocyte PBG deaminase activity.^{4,5} Genetic testing can also confirm the diagnosis and can identify the familial pathogenic variant.

In this case, more information — including the urinary PBG level before the administration of hemin — is lacking. Genetic testing could have confirmed the diagnosis of acute hepatic porphyria (if positive) but would not have differentiated active disease from clinically latent disease. The urinary PBG level can be very elevated during acute attacks of porphyria, especially in patients with acute intermittent porphyria, and may not completely normalize after one dose of hemin. However, the urinary PBG level can be lower and more transiently elevated in patients with variegate porphyria or hereditary coporphyria.^{4,6} This patient had normal urinary PBG and uroporphyrin levels after one dose of hemin, and the reported history of acute intermittent porphyria in immediate family members could not be confirmed. Therefore, although a flare of acute intermittent porphyria was initially diagnosed and treated with hemin, alternative explanations were explored.

HOSPITAL COURSE

Dr. Charles O. Masaki (Psychiatry): By the fourth hospital day, the patient had received daily infusions of hemin, and there had been no adverse reaction leading to the administration of gluco-

corticoids, sedation, or intubation, although such a reaction had been reported to occur during a previous infusion. However, the abdominal pain, nausea, and jerking movements continued despite treatment with hemin. The patient's mood was described as labile. She was frequently irritable and agitated during interviews and evaluations, particularly when questions were asked about her history of acute intermittent porphyria and genetic testing.

The hematology clinic in the United Kingdom where the patient had previously been evaluated and treated for acute intermittent porphyria was contacted for collateral information. The clinic had no record of a patient with the same name; however, it was noted that the clinic had received multiple telephone calls from hospitals in the United States requesting health information about female patients with similar histories of acute intermittent porphyria. The patients typically had different names but the same date of birth.

A psychiatry consultation was requested.

EVALUATION FOR FACTITIOUS DISORDER

Dr. Mladen Nisavic: Concern about deception syndrome — a broad term that encompasses both factitious disorder imposed on self (frequently shortened to factitious disorder) and malingering — can arise in a variety of ways. For example, an unexpected change in the patient's course of illness, inconsistent collateral information, or emerging evidence of a primary or secondary gain may invoke the possibility of deliberate falsification of illness, even when such falsification has not been directly observed. In this case, the incongruent collateral information from longitudinal providers, coupled with an inability to confirm the patient's identity, first alerted the care team to the possibility of deception even before her laboratory tests were inconclusive. The patient did not identify any family members or friends who could confirm her identity or provide collateral information, which increased our concerns.

Factitious disorder is a diagnosis that can be made only after other disorders have been ruled out. The formal diagnosis of this syndrome is rare. Most studies estimate the prevalence of

factitious disorder to be 1%; the condition is most likely to occur in female patients, health care workers, and patients with a history of trauma or major mental illness.^{7,8} Most patients feign symptoms of a medical illness, often a preexisting medical condition, although deliberate falsification of a psychiatric illness can occur. The deception can involve exaggeration or invention of symptoms, induction of illness through ingestion or injection of substances, falsification of studies and medical records, or exacerbation of genuine medical findings through deliberate nonadherence to treatment.

Deliberate and purposeful deception is the key finding in both factitious disorder and malingering, and this finding helps to differentiate these syndromes from other conditions that may be associated with similar presentations, including delusional disorder, conversion disorder, somatic symptom disorder, or borderline personality disorder. To make the diagnosis, it is imperative to obtain evidence of falsification of illness, whether through direct observation, the review of medical records for inconsistencies, the discovery of tampering with laboratory tests or medical records, or confrontation of the patient. If there is clinical concern about deception syndrome, continuous observation with a 1:1 sitter is recommended, along with meticulous exploration of any additional records, including controlled-substance databases and insurance or pharmacy records. Such records may reveal not only overt deception but also some other factors associated with the syndrome, including frequent use of health care services, peregrination (movement across multiple health care facilities, sometimes in multiple states or countries), inconsistencies in basic demographic data, or a history of describing grandiose “tall tales” about previous achievements (*pseudologia fantastica*).

The presence of a clear external benefit — for example, access to controlled substances, securing shelter, financial gain, or avoidance of work, responsibility, or criminal prosecution — is indicative of malingering.⁹ The absence of a clear benefit, in which the feigning of symptoms appears to be motivated by the patient’s desire for attention or to reinforce experiences related to a sick role, is indicative of factitious disorder. Patients with factitious disorder may describe a substantial history of trauma, such as early

childhood trauma (e.g., death or absence of a close family member), sexual trauma, or disrupted attachment (e.g., neglect or abandonment). Screening for early trauma is important but may prove to be practically challenging; the patient may not have had time to form trust with the consulting psychiatrist and thus may be unwilling to provide this information. Differentiating primary gain (e.g., reinforcement of a sick role) from secondary gain (e.g., external benefit) is often challenging in the hospital setting, where patients may not disclose the motivation for their behaviors or may not acknowledge any deception when confronted. As many as 77% of patients with factitious disorder do not admit that they have feigned illness even when presented with objective evidence, and most disengage from providers after such confrontation.^{10,11}

Containment, including involuntary psychiatric hospitalization, can be considered as a therapeutic option for patients with factitious disorder. Potential goals of such an intervention include medication initiation or discontinuation (or both), introduction of the patient to group and individual counseling, and establishment of a safe therapeutic space to minimize any acute self-harming behaviors. Patients frequently decline inpatient treatment; involuntary containment is chiefly considered when there are concerns that the patient’s behavior may pose an urgent risk of self-harm.

Although psychiatric treatment is often warranted, especially for patients with a preexisting history of trauma, more than 60% of patients with factitious disorder decline follow-up.¹¹ There are no recommended specific pharmacologic interventions; individual therapy has been shown to benefit motivated patients. Ultimately, the prognosis is poor, given the increased morbidity and mortality related to feigning illness or undergoing unnecessary medical or surgical interventions.

FOLLOW-UP

Dr. Schell: A multidisciplinary meeting that included the medicine, hematology, and psychiatry services was held with the patient. The information about the lack of collateral evidence from the hematology clinic in the United Kingdom was described to the patient, and the concern

about deception was shared. She elected to leave the hospital, and she was discharged home with no medications.

On the day of the patient's discharge, a woman with a different name and date of birth sought evaluation at the emergency department of a hospital affiliated with this hospital. She reported falling off a dirt bike. An ankle dislocation was treated, and the patient was discharged home. Four days later, the woman again sought evaluation at the emergency department of the other hospital because of symptoms consistent with her previous flares of acute intermittent porphyria. She was admitted to the intensive care unit of the other hospital because of previous adverse reactions to hemin infusions.

A hematology physician who worked at both hospitals recognized the details of the patient history. A multidisciplinary meeting that included the medicine, hematology, and psychiatry services was held with the patient. A photograph of the face from the recent admission at this

hospital, taken after the jaw dislocation, was shown to the patient. Concern about deception and factitious disorder was shared with the patient. Again, she elected to leave the hospital, and she was discharged home with no medications.

During the subsequent months, five separate identities were discovered in this hospital and affiliated hospitals in New England. In addition, this hospital received telephone calls from two other hospitals in the mid-Atlantic region that were requesting collateral information about women with similar details in the patient history.

FINAL DIAGNOSIS

Factitious disorder.

This case was presented at the Medicine Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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REFERENCES

1. Case Records of the Massachusetts General Hospital (Case 12-2014). *N Engl J Med* 2014;370:1542-50.
2. Van Dinter TG Jr, Welch BJ. Diagnosis of Munchausen's syndrome by an electronic health record search. *Am J Med* 2009;122(10):e3.
3. Bissell DM, Anderson KE, Bonkovsky HL. Porphyria. *N Engl J Med* 2017;377:862-72.
4. Anderson KE, Lobo R, Salazar D, et al. Biochemical diagnosis of acute hepatic porphyria: updated expert recommendations for primary care physicians. *Am J Med Sci* 2021;362:113-21.
5. Lefever S, Peersman N, Meersseman W, Cassiman D, Vermeersch P. Development and validation of diagnostic algorithms for the laboratory diagnosis of porphyrias. *J Inher Metab Dis* 2022;45:1151-62.
6. Anderson KE. Acute hepatic porphyrias: current diagnosis & management. *Mol Genet Metab* 2019;128:219-27.
7. Fliege H, Grimm A, Eckhardt-Henn A, Gieler U, Martin K, Klapp BF. Frequency of ICD-10 factitious disorder: survey of senior hospital consultants and physicians in private practice. *Psychosomatics* 2007;48:60-4.
8. Bass C, Halligan P. Factitious disorders and malingering: challenges for clinical assessment and management. *Lancet* 2014;383:1422-32.
9. Case Records of the Massachusetts General Hospital (Case 26-2019). *N Engl J Med* 2019;381:763-71.
10. Krahn LE, Li H, O'Connor MK. Patients who strive to be ill: factitious disorder with physical symptoms. *Am J Psychiatry* 2003;160:1163-8.
11. Eastwood S, Bisson JI. Management of factitious disorders: a systematic review. *Psychother Psychosom* 2008;77:209-18.

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